Zahra Karami

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Education

Science and Research Branch, Islamic Azad University Sep. 2022 – Jul. 2026
Bachelor of Science in Computer Engineering Tehran, Iran

- GPA: 19.37/20 (4.00/4.00)

Areas of Interest

- Artificial Intelligence and Machine Learning
- Deep Learning
- Reinforcement Learning
- Computer Vision and Pattern Recognition
- AI for Healthcare and Medical Imaging
- Software Engineering

Honors

- Ranked 2nd among Computer Engineering students.
- Ranked among the top 10% out of 125 undergraduate students, School of Electrical and Computer Engineering, Science and Research Branch, Islamic Azad University.

Publications

Pixels or Queues? A Careful Evaluation of Deep and Tabular Reinforcement Learning for Traffic Signal Control 2026
Zahra Karami and Azadeh Zamanifar Preprint
Code and data: github.com/zahraKrm/Deep-RL-Traffic-Signal-Control

- Built a CNN-based DQN that sets signal phases from raw 20×20 occupancy images; on the original waiting-time metric it beat tabular Q-learning by 17.2% and fixed-time control by 44.9%.
- Ran a rigorous evaluation on 5,000 held-out episodes with strong baselines and seed-level statistics, showing that all-red clearance ticks hold 31–65% of delay and change the controller ranking.
- Interpreted the learned policies with saliency, occlusion and input ablations, and showed that a phase-aware, clearance-inclusive reward cuts switching by 71%; all results are fully reproducible.

Related Courses

- Artificial Intelligence and Expert Systems (20/20)
- Data Structures (20/20)
- Algorithm Design (20/20)
- Fundamentals of Data Mining (20/20)
- Engineering Statistics and Probability (19/20)
- Databases (20/20)
- Computer-Aided Design of Digital Systems (20/20)

Notable Academic Projects

N-Puzzle (Sliding Puzzle) Solver: A* vs. RBFS | *Artificial Intelligence* Spring 2025

- Implemented and compared A* and Recursive Best-First Search (RBFS) for solving the N-Puzzle using the Manhattan distance heuristic, evaluating computational efficiency based on execution time and expanded nodes.

Pacman AI — Search & Heuristic Optimization | *Artificial Intelligence* Spring 2025

- Implemented informed and uninformed search algorithms (DFS, BFS, UCS, A*, and Greedy Best-First Search) for pathfinding in the UC Berkeley Pacman AI environment; designed Manhattan-distance-based heuristics and state representations for multi-goal corner and food-search problems.

Constraint Satisfaction & Optimization — N-Queens, Map Coloring & Sudoku | *Constraint Satisfaction* Spring 2025

- Solved N-Queens, Map Coloring, and Sudoku as constraint satisfaction problems using Backtracking, MRV, Forward Checking, and Arc Consistency (AC-3) for efficient search and domain filtering.
- Explored Hill Climbing, Simulated Annealing, and Genetic Algorithms as alternative optimization approaches for the N-Queens problem.

Data Mining & Machine Learning on Kaggle Datasets | *Machine Learning & Data Mining* Fall 2025

- Developed ML pipelines across five Kaggle datasets, including preprocessing, missing-value imputation, classification (SVM, KNN, Random Forest) and regression (Linear Regression, KNN, Gradient Boosting), with model evaluation and feature importance analysis.
- Applied K-Means, DBSCAN, and Agglomerative Clustering, evaluating cluster quality with Elbow, Silhouette Score, Inertia, and Dendrograms, and explored dimensionality reduction using PCA and t-SNE.

Food Ordering Web Application | *Web Development* Winter 2025

- Developed a component-based food ordering web application in React, featuring meal listings and an interactive shopping cart.
- Implemented application state management using React Hooks and Context/Provider architecture to manage cart interactions and dynamic UI updates.

Responsive University Website | *Web Development* Winter 2024

- Developed a responsive, interactive university website using HTML, CSS, JavaScript, Bootstrap, and jQuery, with adaptive layouts for desktop and mobile devices.
- Implemented interactive UI components including responsive navigation, tabs, carousels, hover animations, collapsible sections, pricing cards, and contact forms.

Technical Skills

Programming Languages	Python, C/C++, JavaScript, SQL, PHP, HTML/CSS, MATLAB
Machine Learning & Data Science	TensorFlow, PyTorch, Keras, NumPy, OpenCV, Matplotlib, Q-Learning, Deep Q-Networks (DQN)
Web Development	React, Firebase Realtime Database, Bootstrap, jQuery, Streamlit
Tools & Technologies	Linux, Git/GitHub, VS Code, LaTeX, Simulink

Certifications

- Web Design Pack (150 hours), Tehran Institute of Technology April 2024
- PHP and MySQL Web Development (80 hours) Jan 2025
  Coursework: PHP, MySQL, React

Languages

- **Farsi:** Native
- **TOEFL iBT:** 100/120 (5/6)